

Fangu Chen

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Department of Mathematics
University of California, Berkeley
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EDUCATION

University of California, Berkeley

Ph.D. in Mathematics

Expected May 2026

Advisor: Yunqing Tang

University of Michigan, Ann Arbor

B.S. in Mathematics

2021

University of Michigan - Shanghai Jiao Tong University Joint Institute

B.S. in Electrical and Computer Engineering

2021

PUBLICATIONS AND PREPRINTS

- Infinitely many supersingular primes for some Mumford's abelian fourfolds. arXiv:2511.06654.
- Superspecial primes for QM abelian surfaces over real number fields. arXiv:2511.07814.
- Determining optimal test functions for 2-level densities, with Elzbieta Boldyriew, Charles Devlin VI, Steven J. Miller, and Jason Zhao, *Research in Number Theory*, 9 (2023), no. 2.
- The limiting spectral measure for an ensemble of generalized checkerboard matrices, with Yuxin Lin, Steven J. Miller, and Jiahui Yu, *The PUMP Journal of Undergraduate Research*, 4 (2021), 202–221.

INVITED TALKS

- Caltech, Number Theory Seminar, January 2026
- Zhejiang University, Number Theory and Representation Theory Seminar, December 2025
- Boston University, Number Theory Seminar, December 2025
- Shanghai Institute for Mathematics and Interdisciplinary Sciences, Workshop: p-curvatures, October 2025
- AMS Special Session (Hodge theory, algebraic cycles, and arithmetic), St. Louis, October 2025
- University of Michigan, Ann Arbor, GLNT Seminar, October 2025

TEACHING EXPERIENCE

University of California, Berkeley

Graduate Student Instructor

Math 110 (Abstract Linear Algebra)

Spring 2026

Math 56 (Linear Algebra)	Fall 2024
Math 16A (Analytic Geometry and Calculus)	Fall 2023
Math 54 (Linear Algebra and Differential Equations)	Spring 2023
Math 91 (Special Topics in Mathematics Linear Algebra)	Fall 2022
Math 53 (Multivariable Calculus)	Spring 2022
Math 10A (Methods of Mathematics: Calculus, Statistics, and Combinatorics)	Fall 2021

Math REU Mentor Summer 2025
 proposed and supervised research project in Diophantine geometry for three undergraduate students; students completed a research paper

University of Michigan, Ann Arbor

Course Assistant

Math 396 (Honors Analysis II)	Winter 2021
Math 395 (Honors Analysis I)	Fall 2020

Math Lab Tutor Winter 2020

Preliminary Arizona Winter School

Elliptic Curves and Abelian Varieties, Problem Session Leader Fall 2023

HONORS AND AWARDS

- Lehmer Fellowship in Number Theory Spring 2024
graduate fellowship at UC Berkeley
- Stephen Smale Award in Mathematical Sciences 2021
awarded by the LSA Honors Program at the University of Michigan
- Outstanding Graduating Senior Award 2021
departmental award to graduating students who have shown the highest level of academic excellence in mathematics at the University of Michigan
- Wirt and Mary Cornwell Prize 2021
departmental award to a student at the University of Michigan who has demonstrated the greatest intellectual curiosity and given the most promise of original study and creative work
- William Leveque Award in Number Theory 2020
departmental award to an undergraduate who excels in the study of number theory at the University of Michigan